

A STUDY ON THE ROLE OF ITTY ACHUDEN IN THE COMPILATION OF HORTUS MALABARICUS

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Hortus Malabaricus, published from Amsterdam in 1678, is the first book printed on the natural plant wealth of Kerala which included the mythical spices and medicinal plants of Malabar. From time immemorial, the precious spices of Malabar were known through legends in Egypt, Greece, Rome and other parts of the civilized Old World. The European discovery of Asian plants was mainly the result of their quest for a sea route to the homeland of these titled spices and medicinal plants. *Hortus Malabaricus* is also the first ever book in which the Malayalam language and the Malayalam script of Malabar of India appeared in print. For this reason, the people of Malabar (Kerala) have a special affection and a feeling of sentimental attachment with this book.

However, until the book was reprinted and published in 1988 only four or five copies of the original book were available in India and only a very few people actually had even seen it. All the 12 volumes of this book were of double folio size, with the entire text written in Latin. As only few people can now comprehend and understand 17th-century Latin and as the subject matter dealt in it was medicine and botanical sciences, one can imagine how many persons today have studied the book seriously. The botanical, historical and cultural investigations done recently during the past few years (Fournier 1988, 1987a, 1987 b; Meniger 1988, 1988; Maral 1979, 1980, 1981a, 1981b, 1984, 1985) could remove many misunderstandings among the laymen regarding the book and its origin.

The enthusiastic, passionate connection of the people of Kerala to the *Hortus* has resulted in many popular articles on the book appearing periodically in Malayalam magazines and other publications. However, most such articles were containing many mistakes not only about its historical and scientific aspects but even about the physical features of the book that could normally have been seen by all. Ordinarily such slip-ups in the popular articles could have been considered as the result of the excitement and passion of the writer, and ignored as such without criticism. There has been a sudden spate of articles that started appearing from the

mid 1800s, severely distorting the fact, which have been popularly quoted by subsequent writers as reality. It was felt that the facts should be made public so that more confusion could be avoided in future. If such mistakes are not pointed out now, they may be accepted as the truth by others.

Before entering into a discussion on the topic we should remember a related matter. At the time when *Hortus* is written, two books were already available in print in Europe: (1) *Catalogue des Simples et Drogues et Cures Medicinales de l'Inde*, written by Garcia de Orta, published from Goa in 1563, and (2) *Travels de los Drogas y Medicinal de las Indias Orientales*, written by Cristibal Jacorta, published from Burgos in 1578. Both these books together contain general information about a total of 100 native plants.

Physical Features

Physical features are those attributes of the book that can be perceived by all, with no special expertise in science or Latin.

1. Number of pages

Most of the Malayalam writers have written that the 12 volumes of *Hortus* contain 500 pages each, totaling about 6,000 pages. As most writers have repeated this in their different articles, this can not be considered as a mere printing error. Actually the number of pages found in Volumes 1 to 12 of *Hortus* are 112, 111, 88, 127, 121, 106, 113, 88, 121, 130, 134 and 148, coming to a total of 1,524 pages. Apart from this about 92 pages spread in 12 Volumes are used for Foreword, Introduction, Dedication, etc. Thus, the total number of pages come to 1,616.

2. Number of Figures

Malayalam commentators mention the number of Figures differently as 740, 796, 800, 5000, 7040, etc. The total number of Figures actually found to be printed in *Hortus* is 791.

3. The year of publication of the First Volume

Many writers give the year of publication of the First Volume differently as 1675, 1678, 1679, 1680 and 1681. The first edition of the First Volume was published for the first time in the year 1678.

4. The year of publication of the 12th (Final) Volume

Controversies exist in different articles regarding the year of publication of the 12th (Final) Volume of *Hortus*. In most literature the year of publication of the Final Volume is given variously as 1703, 1706, 1705. Till recently it was generally accepted as 1703. However, from recent investigations it has been proved beyond doubt that the 12th Volume came out of press in 1695. A Review of it published in a *Science*

periodical has come to light proving this. Copies of a reprinted Edition of the 12th Volume published in 1903 has been in extensive circulation, which made people to believe this error.

The authorship of Hortus Malabaricus

The main point of controversy regarding this book that has been widely disputed and challenged in the popular Malayalam articles is the authorship of it. In the front page of each of the 12 Volumes of the book it is clearly printed that Hendrik Adrian Van Rheede is its Editor. Although there are some differences in the spelling of his name written in different Volumes, there is technically no doubt that Van Rheede is the author of the book. Also, no one has ever come forward claiming the entitlement of the authorship. When the First Volume of Hortus was published in 1685 itself, many opponents of Van Rheede in The Netherlands had claimed that such a book on plants and medicine can not have been written by some one who is not a scientist or physician but a mere soldier. A fitting reply to such a criticism was given by Van Rheede himself in the Foreword to the 1st Volume published in 1682. There is no need to repeat here the points mentioned quite in detail in that published reply. In the title page of the First Volume itself the name of Van Rheede is given as its "Editor". The names of the famous European physician Theodorus Janoon of Amsterdam, MD and plant scientist J. Commelin are also given as (additional) Editor and Commentator. The sources of information provided in the book are also recorded in the first pages clearly and unambiguously. Therefore, Van Rheede can not be accused of plagiarism either.

The co-operation and involvement of a large number of experts is essential for the writing of such a book. Undoubtedly, the original idea to prepare such a publication, planning of the contents of the book, gathering and coordinating the team of experts, collection and preparation of the diverse materials, arranging for the astronomically huge funds to meet the expenditure to collect the materials and man-power, were all done by Rheede alone. Even when it became a serious threat for the future of his official position and the safety of his life itself, Rheede did not deter from his aim but went ahead with the task of completing his book. Had Rheede did not complete this magnum opus succumbing to the grave challenges, the ancient traditional medico-botanical knowledge of Malabar (Kerala) compiled and preserved in Hortus Malabaricus would have been completely lost. Even the names of the great Malabar physicians who helped in the compilation of this knowledge would have been totally erased from the history of Malabar and India.

The names of many people who are involved with the writing of Hortus

are recorded in the book. One such name is of a priest Fr Mathew of St. Joseph. Many articles have been appearing in the Malapalam publications stating that, as the person who helped in compiling all the information presented in the book, Fr. Mathew should be considered as the actual author of *Hortus Malabaricus*. One writer say that the entire book was written by Fr. Mathew and, in order to get into the favour of the Governor, the name of Van Rheede was inserted as its author and the completed manuscript of the book was presented to him, to publish as his own. He has boldly gone to the extent to say that "some people even shout that the said Van Rheede is the real author of the said book". Such statements can come only from people who have not even seen the *Hortus*.

European Collaborators of Hortus Malabaricus

Apart from the translator and interpreters, the names of the following European experts are mentioned in *Hortus Malabaricus* in different volumes as Rheede's collaborators. Among them the name of Fr. Mathew is not given anywhere.

1. J. Commelin, Vols. 1-12.
2. Jansen ab. Kinslowen, Vols. 1-6.
3. Johann Camerlinus, Vol. 3.
4. Johann Munnicks, Vols. 3-5.
5. Abrahamus Paet, Vols. 7-12.
6. Willem ten Rhine.

The first five of them were working in their own parent institutions in Europe and were reading and editing the manuscripts and illustrations sent to them by Rheede and were readying them for publication. When Rheede once met the Dutch expert Willem ten Rhine at Batavia, he requested for the help of ten Rhine to go through and correct the descriptions of the plants, that he agreed to, seen in some literature (Petiver, 1698; Herpin, 1988). No other contributions were made by any of them in the compilation of *Hortus*.

Fr. Mathew of St. Joseph

Although his name is not found to be included among the authors, Editors or interpreters in any Volume, the name of Fr. Mathew is mentioned by several writers (of widespread common articles written in Malayalam and English) more prominently than the actual author van Rheede himself, the matter has to be examined carefully and in some detail.

It was based on his firm belief that it would be advantageous for his Government as well as his Company (VOC) that Van Rheede took the decision to compile such a book. When the Dutch officials and soldiers

in Cochin garden fell ill, they used to report it to The Hague and get the required medicines for their treatment from there by ship. Being a sharp observer of Nature, Van Rheede noticed that such medicines coming here were similar to the medicinal plants growing naturally locally. These medicinal plants were collected by Arab traders from here and sold in European markets by them and were coming back to Malabar. The correct source land of them was not known to others as that was kept as a highly guarded trade secret. Since they were brought and sold by Arabs there, it was known in Europe (Mediterranean ports) as Arab medicine.

From his investigations Rheede also learnt that the people of Kerala (Malabar) were using the local medicinal plants to effectively treat their illnesses. Medicine was a highly developed science in Malabar, with proficiency available locally. Without any doubt, Van Rheede came to the conclusion that if the information about the valuable medicinal plants of Malabar is compiled, it would be highly economically and politically advantageous for the Company. The sharp intellect of Rheede would also have recognized the immense possibilities of diversifying the trade that till then was mainly revolving on Pepper. He also realized that the information would be a major endorsement for his opinion that Cochin of Malabar is more suitable to be the second eastern capital of the Dutch Company instead of Colombo of Ceylon. It was purely incidental that *Portus Malabaricus* became an asset to the world botanical science.

Van Rheede was a keen observer of nature. He was confident such a book would bring immense economic benefit to the Company. However, he accepted that neither he himself nor any other employees of the Company had the time or expertise or technical know-how to write such a book. The best alternative is to entrust the work to someone learned in botanical and medical sciences. His hunt for such a suitable man led him to the information that there is a Catholic priest named Fr. Mathew who is learned in plants and medicine was residing in the neighbourhood of Cochin, under the jurisdiction of the Raja of Cochin. Therefore Rheede made arrangements to bring Fr. Mathew to meet him immediately.

Fr. Mathew's earlier name was Pietro Pigna. He was born in 1617 in a small town Marcellino, near Naples. After studying Medicine in Naples University where he is said to have taken a doctor's degree, he joined the Discalced Carmelites missionaries in 1639 accepting the monastic name Mathew of St. Joseph. In 1644 he was sent to the East for missionary work and spent four years in Mount Carmel in Palestine. Cosentinus of St Columa was his superior there. Cosentinus was the brother of Jacob van Gool who was the Professor of Oriental languages in Leiden and was interested in Botany. Van Gool had entrusted Fr. Mathew to send him

information on local plants and their sketches, whenever he visits east. As desired by the brother of his superior, Mathew had sent an account of the plants of Lebanon and sketches of some of them. In 1546 Mathew was transferred to the Mission to Persia and the Indies and worked in Basra of Iraq. His missionary work among the Muslims was not successful. In Iraq he learned Arabic but Mathew became convinced that working among the Muslims on medical care is more successful and respected than religious conversion. While at Basra, he wrote a book on medicine and hoped the Jesuits would print it but it was never published. He had also copied the illustrations of many local plants drawn by Saladin Artale during his stay in there (Heniger, 1986).

Mathew left Basra in 1551 and settled in Diu, a Portuguese possession on the coast of Gujarat in India. From there he undertook some short journeys to East Africa for missionary work. There too he was taking notes on local plants and their medical uses and drawing their rough sketches. His collection of notes and drawings which was known among his colleagues as *Vindiarium Orientale* became quite large. His brief notes were containing the (local) names of such plants and to treat which illness they are used. Their rough drawings are made by pen to help him remember the concerned plants and not as an outcome of skilled art. A part of this *Vindiarium Orientale* was included in their own publications by Zanoni (1575) and Monti (1743).

In 1555 Mathew was asked to go to Malabar, where the Discalced Carmelites had an important missionary province known as the Church of Serra at Ankamash (Syro Malabar Church), where also he continued to study the local plants. Mathew was sent to Serra in connection with the wish of Pope Alexander III to reconcile the Roman Church with the Malabar Christians of St. Thomas and to reverse the influence the Syrian Patriarch. The apostolic commission ultimately succeeded in asserting the authority, and on 23 July 1557 the Catholic Christians of St. Thomas took the oath of allegiance to the Roman Church (Gustache di S. Maria, 1718; Heniger, 1986). Guiseppe di S. Maria was proclaimed as bishop of the Church of Serra, who mentioned that Mathew was his companion.

The position of Carmelites in Malabar became threatened when the Dutch laid the first siege to Cochin during November 1661-March 1662. This siege was not successful and the Dutch had to withdraw. During the second siege Cochin was capitulated on 7 January 1663 and bishop Guiseppe and other priests who were there fell into the hands of the Dutch. Mathew escaped as he was not present there at that time. According to the capitulation treaty all the Roman Catholic priests of European origin had to leave Malabar (i.e., Cochin) and Guiseppe and

others were banished from there by the Dutch Admiral Van Goens. When Giuseppe protested referring to the papal commission letter, Van Goens remained adamant and stated that the Malabar Catholics are 'amply provided with their own priests and are quite content with them'. On 1 February 1663 he consecrated the Malayalee, Alexander de Campo (Paramel Chandy) as the bishop of the Church of Serra. This was how the first native Indian Bishop was ordained. After three days on 4 February 1663 Giuseppe was sent by ship to Goa. Realising that he will not be allowed to return to Cochín, on 22 January 1664 he left via Surat and Bassa, reaching Rome on 5 May 1665.

During this period Fr. Mathew was residing outside the jurisdiction of the Dutch. He accompanied Giuseppe when he returned to Rome, taking most of his manuscript notes and sketches of *Historium Orientale*, to try to get it published from there. Although some parts of it was included in a publication brought out in 1675 by Giacomo Zanoni (Milan, 1740), *Historium Orientale* containing the brief notes and rough sketches of plants of the Eastern countries that Mathew visited, including several plants of Malabar, was never published as a separate book.

Fr. Mathew then returned to Malabar clandestinely and, to escape the wrath of the Dutch was careful not to come in contact with them or ever appear before them. Nevertheless he was known locally as one with a fair knowledge about the plants of Malabar. Therefore when it came to be known that Rheede was on the look out for a man to help him in the preparation of a book on the medicinal plants of Malabar, the name of Fr. Mathew was suggested to him.

Knowing that it is impossible to get sufficient time or leisure in between his official duties for the preparation of a book on sciences, Rheede made arrangements to bring Mathew to him and discussed the matter. The suggestion was attractive to Mathew. Under normal circumstances he would never have been allowed or permitted to enter Malabar region and the vast official facilities and support which ordinarily would never imagined to be made available to him, were all offered to him in a platter. Mathew accepted the request for this help.

Rheede (1682) writes about his interview with Mathew: 'I learned that under the sovereignty of the King of Cochín there lived a certain man endowed with the greatest possible knowledge of Indian plants, which he had drawn from nature. This man, whom I have mentioned just now, was a very learned man of sixty, living a sober life, by profession a pontifical priest of the order of the Discalced Carmelites; his name was Father Mathew of Saint Joseph, a born Neapolitan. He had lived for many years in Arabia and Persia and at that time had been sent a future

confessor to the Archbishop of the St. Thomas Christians of Malabar. When summoned, this venerable man did not hesitate to come to me in the town of Cochin and, when he had understood my purpose, viz., that a list and description of the plants of Malabar had to be made, for which I promised this venerable father my assistance, he very kindly and promptly offered me his services in turn. At the same time he showed me some rough sketches of plants which he had formerly drawn as an aid to his memory with pen and ink, viz., one leaf, one flower and one fruit of each plant, of which he had noted down in particular the curative virtues, with the aid of which this pious old man had helped, on his journeys, not only himself but also very many others. Thus the first beginnings of this *Hortus* were that from the first sketches of plants, copied on other paper, arose a more accurate delineation of them through the care of Frs. Father Mathew, who, relying solely on memory, turned out the natural forms of the plants in a wonderful way from the rough sketches."

It was after getting a definite promise of help from Mathew, Rheede commenced his plan for the compilation of the book.

In this arrangement two things have to be remembered clearly. (1) The first beginning of *Hortus Malabaricus* was from the notes and sketches made by Mathew. (2) *Vindiarum Orientale* of Fr. Mathew was not the starting point of *Hortus Malabaricus* because, (i) its manuscript, or most of it, had already been sent to Europe to try for its printing and publication, (ii) the brief notes and rough sketches made for his *Vindiarum Orientale* were originally to prompt his own memory. (iii) only a few plants of Malabar were dealt with in *Vindiarum Orientale*. Heniger (1966) writes: "In fact *Vindiarum Orientale* was no more than Mathew's medico-botanical notebook for the consultation of relevant information on the medicinal plants during his travels". Van Rheede has also recorded: "The (Fr. Mathew) had noted down in particular the curative virtues with the aid of which this pious old man had helped, on his journeys, not only himself but also many others". Though some parts of *Vindiarum Orientale* was published by some other authors in their publications (Zaneri, 1671; Monti, 1792; Saccardo, 1894), it was never published independently as a book. It may also be remembered that two other books, one by de Cota (1563) and another by d'Acosta (1576) were already available in the book shops in Europe in those days.

Rheede's help to Fr. Mathew to pursue the latter's ecclesiastical duties

Rheede provided all support to Fr. Mathew for the preparation of the book. Rheede also helped him to construct a church at Ernakulam of

Malabar so that he can attend to his ecclesiastical duties as a catholic priest without any hindrance and frequently visit Cochin Fort. This is how the first Carmelite church at Chathiah was founded. Later, exploiting the goodwill of Rhoads, Mathew also built another church for Catholics at Wangapoly. Rhoads wholeheartedly concealed his goodwill towards Mathew, for the support and building of Catholic churches was decidedly not in keeping with the Company's policy (Heriger, 1986). For the sake of success of his project on *Hortus Malabaricus*, Rhoads was taking a dangerous risk. However all these arrangements did not succeed as expected by him (Fournier (1987:4).

The academic efforts of Fr. Mathew were not in accord with the high standards expected by Rhoads. Cassius (1678) writes: "the sketches of plants, hurriedly drawn by Mathew and their descriptions were based on his memory and imagination. But, the respected Lord (Rhoads), and I myself, feel that they have to be improved much. Their medicinal effects were also noted. Their descriptions were superficial. As for the sketches, the shape of plant parts, especially flowers, fruits and seeds, were not at all accurate". Heriger (1986) has also commented: "The sketches of flowers and fruits are done without any detail so that their sketches are not good enough even to make a beginning for their identification".

While writing about the initiation of *Hortus Malabaricus* Van Rhoads (1682) has written in Dutch: *Hortus Indiarum* Volume 1: "Thus, the first beginning of the work was from the rough sketches made from his (Mathew's) memory. Because he delineated everything with a pen, and that only in outline, being inexperienced in painting with a brush, the pictures were not very much like the living plants. Moreover, the minor parts of the plants could not be shown accurately, or at least not distinctly, so that the true forms of the living plants could not be recognized at all these places, especially in those pictures made with a pen neither light nor shadow could be shown somewhat accurately. Of course the excellent old man had not imagined that it would be necessary to greater and more accurate pains in depicting and describing those plants, because he himself with the aid of his memory was indisputably acquainted thoroughly with the plants presenting themselves to him and their curative virtues. Moreover, these rough sketches had nearly always been made on his journeys, so that fairly often now the fruit of one plant, now the flowers of another plant were missing, and sometimes both. Nevertheless I corrected these defects as well as I could by seeking precisely for those plants and supplying the deficiencies. But still many plants have not been heard, either because he had not noted down the names accurately, or the venerable old man had received them from men who did not know the common names of the plants by which the botanists of Malabar designate each plant, or because they were plants from another climate. These approximately were the

obstacles, which disturbed the beginnings of *Hortus Malabaricus*. Among them the greatest was that if you compared the drawings with the plants themselves, you would hardly have said the pictures had been made from them, so poorly did they accord therewith. As to the descriptions, this often cohered so little through perfunctory and fleeting attention that it might have referred to several plants, even if they were quite different".

In brief, the descriptions and sketches being made by Mathew for *Hortus Malabaricus* were entirely unsatisfactory and useless. "At this time, there arrived in Malabar from the island of Caylon the famous Paul Herman, Doctor of Medicine, who had just been appointed Professor of Botany at the illustrious University of Leiden. When he had seen that the beginnings of our *Hortus* were impeded by so many obstacles and by the rather inaccurate nature of the descriptions as well as the pictures, with modest kindness he saw fit to give us to understand that in this way it would hardly become a book of any importance. Nevertheless, the shortness of his stay with me was the cause that I could not discuss my collection of the *Hortus Malabaricus* with him, neither could he assist me with wise advice in this very complicated work. Circumstances being such, I wanted to begin the whole enterprise anew, and I devoted myself to describing the plants as well as I could and working at this *Hortus* in another way" (Rheede, 1982).

Paul Herman said that Mathew was not writing any scientific book (Fournier, 1988; Herzig, 1988). Van Rheede realized that Mathew's knowledge of the flora of Malabar did not go much beyond what he needed in the daily practice as a priest and a physician. A new detailed plan was suggested for writing *Hortus Malabaricus*. Herman's judgment convinced him in his resolve to take the assistance of local Malabar scholars for this project. Local experts will decide which plants are to be included in the book. Fresh specimens of these medicinal plants, with the leaves, flowers and fruit, will be collected locally from Cachin and nearby localities of Malabar from nature. Their correct local names known to the Malabaris, their medicinal and other important uses will be described to Rheede by the local experts with the help of his official interpreters. Three or four painters, who stayed with Rheede in a convenient place, at once accurately depicted the living plants readily brought by the collectors. To these pictures a description was added nearly always in his presence (Rheede, 1982; Herzig, 1988).

Among the interpreters two persons, Ity Achuden and Muzysk Parath were conversant with Portuguese language. The official interpreter of the Dutch Company Inmanuel Carneiro helped to translate Malayalam into Dutch. Inmanuel Carneiro and Herman Christian de Doney translated from Portuguese to Dutch and Herman Christian de Doney and Johann

Casertinus from Dutch to Latin. Another Dutch interpreter Peter Minnes also helped to translate some parts from Portuguese (Heniger, 1988).

The most significant decision taken by Rheede at that juncture was to totally reject all that had been done till then by Mathew, as it was far from the high standard expected by him (Poumer, 1987a). He decided to do the work with the help and collaboration of local experts. Their contribution in determining the contents of the book was decisive. It was they who decided the plants to be included in the book. They collected the plant specimens and introduced their names and medicinal virtues to Rheede. Most importantly, all the details of these plants given in the book were supplied by the local experts of Malabar.

Painters and Engravers

Almost all the sketches drawn by Mathew were made by writing pen during his travels. As he had no training or experience in drawing with paint and brush the different plant parts were not drawn properly. The sketches made by him were merely to aid his own memory. On many occasions Rheede and Herman who saw these sketches have commented on the inaccuracy of the sketches and their undependability. They have no resemblance with the concerned plants. A comparison between Mathew's sketches made for his *Vindiarum Orientale* and the original drawings of Rombus Hildebrandt still preserved in London will convince anyone about the huge technical and artistic differences between them.

At this point it may not be inappropriate to say a few words on the details of *Vindiarum Orientale*. As mentioned earlier, *Vindiarum* is a collection of brief notes and rough sketches of medicinal plants of middle and near eastern countries visited by Mathew. Most part of this was brought to Italy by Giuseppe from where it reached Celestino of St. Ubaldo. Later Celestino who was appointed as the Professor of Oriental Languages in Rome added about 600 sketches drawn by himself to this collection but they were never published.

Another part of *Vindiarum* reached Michael de Byrio, who sent it to Giacomo Zanoni. A third small portion was also sent by Byrio to Zanoni through Valartous of St. Joseph. Descriptions of about 50 plants mentioned by them were included in his book *Historia Botanica* published in 1675 by Zanoni. A Latin edition of it in the name *Parlorum Virgipum Historia* was published 1742 by Gaetano Lorenzo Monti. Most of the sketches drawn by Mathew, including those redrawn by Celestino, are preserved in the Museum of Natural History, Pisa. These collections known as Pisa Codex contains the descriptions and sketches of many plants found in Cyprus, Palestine, Lebanon, Syria, Persia, Mozambique,

India and Caylon Zanoni and Sebastian (1794) have recorded that these sketches were made by writing pens. Some of them were copied from Saladin Arafah made in Java. They are beautiful drawings but, as they are not drawn showing details it's not possible to identify the concerned plants from them (Heriger, 1986). When you compare the actual plants and the sketches they represent, you will never believe their identity; they are so poorly represented (Rhede, 1982).

The beautiful and technically accurate sketches printed and ultimately published in *Hortus* are drawn by experienced expert draftsmen appointed by Van Rhede after those discarded by Mathew. Rhede has recorded that three or four draftsmen were staying with him to draw the sketches. The names of two of them are available from the Dutch records. One was Antoni Janits-Goediker, a member of the Dutch Garrison at Cochin. He was a member of the famous family of artists of Antwerp city with the same family name (A. van Muthach, 1906). He had the rank of Sergeant at that time. Goediker's signature is found in the Fig. 1 of Vol. 1 of *Hortus*. The second draftsman appointed by Rhede is Marcellis Splinter. Below the Fig. 38 of Vol. 6 (in the original copy) the signature of Splinter is found to be given. Splinter belonged to the family of painters and sculptures of Utrecht city who were decorating the Cathedral there. Born in Amsterdam, he was an employee of the Dutch Company at Cochin at that time (P. Swellens, 1925). The names of other draftsmen appointed by Rhede are not known. However, it is doubtful that the artistically and technically perfect drawings made for *Hortus* have nothing to do with the crude attempts of Mathew.

The drawings of plants, etc., copied and engraved in copper plates for printing were done by experienced block-makers of the Netherlands. The names of two of them are available from the Dutch records. Bastian Stupendahl born in Amsterdam and Gonsalves Appelman born in Amsterdam and residing in Antwerp (F. G. Keller & H. A. Gwinibell, 1938).

The portrait of Van Rhede printed in *Hortus Malabaricus* was drawn by the famous painter of Amsterdam, Peter Stevens van Gend (Thieme & Becker, Vol. 5, 1912; Heriger, 1986). A recent investigation (Manilal, 2018) has revealed that the frontispiece drawing of *Hortus Malabaricus* was made by the Dutch Golden Age artist Gerard de Lairesse of Amsterdam.

From the above facts it is clear that Mathew had no role in the preparation of *Hortus Malabaricus* which was finally printed and published. The making of it was solely based upon the co-operation, knowledge and expertise of native scholars and specialists. Although the rough notes and sketches provided by Mathew was its beginning, after the second start his role was restricted to that of general assistance (Heriger, 1986). This

not relate these things, as may be thought, to disparage the work and the exertions of the most honest Rev. Father Mathew, rather on the contrary. Indeed, we owe this man, who was leading a very honest life of true piety, the greatest possible gratitude, because he assisted us so honestly, faithfully, and with such unflinching exertions. Nay, we rather mention this excellent man in order to show that we venerate his memory and shall always do so, and renew the pleasant recollection of his friendship and goodwill towards us" (Rheede, Hort. Malab. Vol. 3, 1682).

The history of Mathew afterwards in Malabar is also noteworthy. As the first associate of Hortius, Van Rheede had a special affection to him. It is not known whether there was any other reason for this friendship. But, none of the other officials of the Dutch Company working under Rheede liked this special companionship, as specifically mentioned in a letter dated 2nd June 1675 by Geiner Verburg sent to the authorities in the Netherlands. "He is said to be an expert doctor of medicine and a good herbalist; that is why he has all the more opportunity to ransack every hole and manage to worm himself so much into the good graces of the Commander Van Rheede".

As the activities of Mathew were so much disliked by Dutch Company as well as the Roman Church, with the permission of the Company, in 1676 the Pope sent a Commission of eight members to Malabar to set right the affairs of the St. Thomas Christians in Malabar. The main issue was to deal with the person of Mathew, who was a constant head ache to them and according to them was "not behaving as proper to a religious priest as was seemly and has refused his superiors the respect and humility he owed to them". The Commission wanted to appoint another coadjutor of Bishop Alexander. Rheede obstructed the election of coadjutor and, to protect Mathew, expelled the Commission from the territory of the Company. While the Pope himself had dropped the priest, Rheede was protecting Mathew, stating that he was "a special friend of the Dutch nation, a merry and kind companion" (Heniges, 1966). Later, when Rheede was transferred from Cochin to Batavia, Rheede instructed his successor Jacob Lobb to keep Mathew in the pay-roll of the Company.

Early in 1678 the Commission of Carmelites suddenly reappeared in Cochin with the intention of removing Mathew. When in Mathew was officiating at night in his small church at Palluruthi, they dragged him like a criminal into their boat and took him to Marapok. Rheede had meanwhile left Batavia and could no longer do anything to help Mathew. Having totally fed-up with the affair of Mathew, the High Government instructed the following commander Martin Huysman to settle it definitely. Mathew traveled empty handed to Surat, disappearing from the history of Malabar

(Classa, 1919). According to Ward (1942) he died in Tatta in the province of Sind, while Chick (1926) states that he died in Cochin where he returned clandestinely and is buried in Knapak (in 1897).

Local collaborators

The names of the following five local persons mentioned below are found to be mentioned in *Hortus Malabaricus*: (1) Vira Kerala Varma, (2) Three Konkani/Jeshmins- Ranga Shat, Vinayaka Parath and Apa Shat and (3) Jity Achuden.

1. Vira Kerala Varma

Vira Kerala Varma. As the Dutch had helped a lot to install Vira Kerala Varma as the King of Cochin, he was indebted to the Dutch. The King was providing all help and assistance to Rheede to undertake his frequent travels to collect medicinal plants mentioned in *Hortus Malabaricus*. A large number of helpers were needed to undertake such travels. 'This party [of collectors], which often consisted of two hundred men, had collected plants there for some hours, they returned, laden with plants they had collected from here, there, and everywhere, to our war ships'. The story of Vira Kerala Varma is recorded in detail in the history of Kerala. After Rheede left Malabar, Vira Kerala Varma entrusted all matters concerning the governance of the state to his old friend Peringalan. However Peringalan looted the treasury and instigated a war with the Zamorin of Calicut which invited the severe displeasure of the people. In 1664 a civil war against the King started, to quell which a military intervention by the Company had to be requested. Vira Kerala Varma became seriously ill in 1663 and died in 1667.

The Foreword written by Van Rheede in Vol.3 praising Vira Kerala Varma (Nothari-Varia, King of Pedirickappe-Souramari, Cochin, Peristeyde, (Ilerta-Sourawan, Cochin, Etc.) is reproduced here:

"Whenever I observe Malabar plants it seems to me I am given an opportunity to make a collection of them with a view to depict and describe them. I often keep a memory of these plants and they make me think how useful it would be to make a collection of them. It is Your Majesty's name that above all occurs to me, you being the Lord of the huge Malabar Gardens. The memory of the benefits I have received from Your Majesty and your subjects and neighbours gives me sufficient strength to carry on the pursuit of my work of making collection from the immense number of trees, fruit-bearing plants and herbs. Encouraged by such recollections when I look back on the period of former days I spent with you, I find how change of fortune struck you. My good days

with you suffered a turn when you were forced to go into exile and seek solace from others, while you were providing me the solace I seek. The dignity of your memory, your outstanding example, brings out the signal way of carrying on the responsibility of the reign entrusted to you. How exalted and worthy of your nobility it is that you could carry on the reign of the kingdom for two thousand and fifty years in succession, subjugating the Indian Kings. Then I remember what change-of-fortune took place. The enemies adopting occult and open ways of attack expelled you from the highest position of a ruler. Your own subjects turned out to be your enemies and forced you to go into exile. Although fortune turned against you, your admirable constancy in the midst of misfortune was kept up in such a way that neither the attack on your honour nor the deprivation of wealth affected Your Majesty. Before long your exalted pristine position was restored; you were crowned King of such vast Kingdom again. I had the fortune to witness all the happiness from the time when Your Majesty was cast into darkest calamity and adversity, and for a while without any hope of remedy. Then dawned the bright days of triumph. With the help of the Belgians the enemies were subdued, triumphantly ascended the seat of the dignity of the Kingdom. At that time I was fortunate enough to serve Your Majesty for seventeen years. In gratitude for all I have received I am impelled to make it known that I am indebted to pay respect to Your Honour during the whole course of my life. Whether present or absent, whether living in diverse places, I do not want to be free of this obligation. It is for the above reasons that the illustrated work of Malabar plants is offered to Your Majesty. Thus, attached to the name of Your Majesty it will spread throughout the world; Your Majesty will be counted among the Dukes and Princes of the Republic of Macaronesia”.

It is observed that the involvement of VasuKerala Varma with the book is limited to his sanctioning, as the King, the full official arrangements to collect the information and plant specimens for the book from his people in Malabar.

2. The three Konkani Brahmins- Ranga Bhat, Vinayaka Pandit and Agre Bhat

Ranga Bhat, Vinayaka Pandit and Agre Bhat. These three were Konkani Sanskrit Brahmins who had migrated to Cochin a couple of centuries back from Goa. The main occupation of these Brahmins was priesthood in their temples that were almost exclusively catering to the people of their own caste (social group). Till recently the western authors were mistaking these Konkans for Namdevadits who are the only indigenous Brahmins in Kerala (Manjeri, 1988: 43).

In the statement given by them in the first Volume of *Herbari Malabaricus* it is mentioned that they belong to the religion of Brahmins and ancient Gymnosophists, residing in the District of Cochin and that they were co-operating with the Noble Lord Henry A. Ribeiro, Governor of Malabar Territories and of the City of Cochin from the year 1634 to 1675. Till recently it was thought by everyone that the language used by them was Sanskrit (Mazhal, 1990) but the information given by them was in Konkani language. One of them, Vinayaka Pandit, also worked as translator from Konkani to Portuguese language. (Excepting this, the names of these three Konkani are found to be mentioned together and no individual information about any one of them is known). A certificate provided by the three of them together translated from Portuguese into Latin by Christian Herman de Doney, and another in Konkani language in Nagari script signed by them are also provided in the First Volume. Though it is not stated specifically, the certificate in Nagari script was possibly written by Vinayaka Pandit. As wanted by Van Rheede, they were in Cochin for two years and conveyed the information on the medicinal powers available in their book "Wanhuwangatnam" to which whatever they have also learned and observed by their own long experience [7]. The plant specimens were brought from the Malabar region to the above mentioned city, "through their servants" who have knowledge of those plants. (Refer the original certificate in the Nagari script). In fact the three were residing in a locality of Cochin within a couple of miles (comfortable walking distance, at a period of time when road transport vehicles were non-existent) from the Dutch Fort where Van Rheede was living.

Some points in the above statement need our special attention here. It is mentioned that the information provided by the three Konkani Brahmins are from their book "Wanhuwangatnam". Ever since I have started my studies on *Herbari Malabaricus* in 1964, I had been trying to find the book "Wanhuwangatnam" but none of the five hundreds of men (including well known Konkani scholars like Prof. Panchabhama Mallappa of Cochin), I contacted or interviewed for this matter in the subsequent decades knew or had heard anything about a book of that name. Then in 1990 an incidental occasion arose that threw some light on the matter. The word Wanhuwangatnam came up as possibly a distorted version of the word 'Maha Nighandu'. When the Sanskrit word Mahanighandu was pronounced by someone whose mother tongue is Konkani, heard by a European knowing neither Sanskrit nor Konkani in 17th century, written down in Portuguese and later rendered in to Latin, such a misrepresentation in Roman letters is understandable. Uncommon words in Eastern languages are not easy on the ears of Europeans.

"Maha Nighanda", meaning "Great Lexicon" is the name of a medical book in Sanskrit language, that was famous in those days. It was also known "Madanapala Nighanda" or as "Madana Vinoda". (As the topic of the role of Konkani in writing the book *Hortus Malabaricus*, has recently made to become a controversial point, please see the separate-General Notes on Konkani Brahmins and Mahanighanda, etc. given below). Since the physicians in Malabar were having their own authoritative medical books written in their own local language Malayalam, Mahanighanda written in Sanskrit was neither well known nor was there a need of its use here. However, that book was considered as an authoritative, popular medical book in the regions north of Kerala, including Goa which was a centre of the Konkani. During their migration to the south the Sarawati Brahmins settled in Konkani (Konkani) were carrying copies of this book with them for their own medical consultation. The role of the Konkani Brahmins, Kanga Bhat, Vinayaka Paraji and Apa Bhat, in the compilation of *Hortus Malabaricus* was mainly restricted to comparing the medical information on the plant species selected by Wil. Achuden for inclusion in the book *Hortus Malabaricus*. (Ref. Ja Rhoads, 1981; Heniges, 1988, 1998; "Local experts will decide which plants are to be included in the book... Their correct local names known to the Malabaris, their medicinal and other important uses will be described to Rhoads by the local experts with the help of his official interpreter"; [d] Fournier, 1980; "He [Rhoads] decided to do the work with the help of local experts. Their contribution in determining the contents of the book was decisive").

General Notes on Konkani Brahmins, Maha Nighanda, etc.

Konkani Brahmins are also known as Gowda Sarawati Brahmins settled in Konkani region. The Sarawati Brahmins are believed to be the Aryan migrants from Central Asia who came to the Indian sub-continent through the Hindu-Kush mountains and the Khyber pass in about 2000 B.C. and settled along the banks of Sarawati River, and hence called Sarawatis, leading an agrarian life. Aitken (geological) period the major rivers of north-west India were Sarawati (Sindhu (Indus), Shatadru (Ganges), Vipasa (Beas), Vitasa (Chenab), Parusni (Ravi), Asikni (Chenab), Yamuna, Ghagghara and Lavanasti. Basically, they were parts of one general channel from Himalayas meeting the Arabian Sea at the Rann of Kutch. There were about 300 cities, and many supporting towns and villages along the banks of Sarawati. It was called the Sarawati-Sindhu civilization. The river course was dotted with lakes and ponds. It was on the banks of Sarawati the Vedic ashrams, that were the centers of Vignedic learning thrived. The goddess of learning Sarawati, therefore, was originally a personification of this river (Ref. Radhakut Hymn in Rigveda 10:75).

After existing for a few millennia, the volume of water flowing down Saraswati began to get reduced and the river did not make it up to the sea. The river finally dried up completely in about 1500-1000 BC. The scarcity of water and the general aridity of the land finally forced the people living on its banks to migrate. The Saraswati-Sindhu civilisation, however, did not vanish. After the economy around Saraswati river collapsed, people moved to east to the Ganga-Yamuna plains, west, northwest and south to Godavari plains. The migration happened slowly, spread over centuries. Due to an acute famine lasting for about 15 years, the last exodus from the Saraswati river banks happened in about 350 BC. Many came to south.

The migrants became widely dispersed, settling along the west coast including the Konkan among other regions. [The Sahyadri Mountain range ("Western Ghats") forms the eastern boundary of the Konkan, and the Arabian Sea marks the western boundary. The southern boundary is the Konganall River. The Mapusa River forms the northern boundary]. Those who live in the Konkan region are known as Konkani. The Gubbi, Gouda and Kurdi tribes are among the original settlers of the migrants in the Konkan region. Gouda Saraswat Brahmins are those belonging to the original Aryan migrants from the Saraswati river banks, who belonging to the Gouda tribe of Konkani.

The migration of Konkani Brahmins to Kerala

The migration of Saraswat Brahmins from Konkan to Kerala mainly occurred in two phases. The first was in the 13th century (midst of 12th AD) and the second in the 15th century (in 15th AD). They sailed south from Goa, along the Konkani coast disembarking at several parts on the way. Many of them came to Calicut, but were not welcomed by the Zamorin, the chief ruler of Calicut and other regions of Malabar, who had a pact with the Portuguese. Therefore the migrants proceeded further south. A small group settled in Cheral in the Yypeen Island, opposite the Cochin harbour entrance, in 15th AD, installing their idol of Lord Parashu brought from Goa. A large group of 368 families of Saraswat Brahmins arrived in 1664 and settled in Cochin with the permission of the Raja of Cochin.

Some relevant information about the Gouda Saraswat Brahmins (GSB) of Cochin and their background found in their official website (gskerala.com) is copied below:

"The social life of GSBs was inseparable from the temples and social exchanges with the locals was limited".

"The GSBs never really integrated to the social structure and lived in fringes of the Kerala society. The GSBs of Kerala never enjoyed Brahmanic

privileges, and never really became part of the larger Indian community which consisted mainly of their Malayali, Tulu, and Tamil counterparts, preferring to maintain autonomy at the expense of integration!

"Two centuries of introverted life has somewhat taken a toll on the Cochin GSIs and has deprived the community of social and linguistic skills required to thrive in a state which has a different social structure than the state in which they have their roots. It has resulted in a community which is essentially introverted which internalises its energies, and is not integrated completely with the mainstream of Kerala society".

"Unfortunately Sanskrit scholarship is non-existent among Cochin GSIs. Cochin GSI priests are only competent in performing pujas, none of them are known to be Vedic scholars. GSI priests officiate mainly in GSI temples and households. Their services are also sought by expatriate north and western Indian communities".

Their observations on themselves of two centuries back (now over three centuries) have not undergone any major change even today. Some of the pertinent statements copied above (within quotations) will be discussed briefly at the end of this Note, for clarification of relevant points.

Maharisingottam or Maharighanda

Maha Nighantu (written as Maharisingottam in introductory pages of *Manu Mahabharata*) literally means "The Great Lexicon". It is the name of a medical book in Sanskrit language. It is also known as "Madanapala Nighantu" or as "Madana Vinoda". The author of this book, Madanapala appears to have been a king and a great patron of learning and is supposed to have flourished about the middle of 14th century A.D. Some of the works ascribed to him are known to have been written by scholars enjoying his patronage. For example, the *Madana-purjaya* was compiled by *Vivekanandacharya* a high authority in the *Benares School of Hindu Law*. The book *Madanapala Nighantu* is designed to be a handy pharmacological glossary, containing names of drugs that are well-known in their country and giving details of properties of those drugs. The detailed information has been provided with respect to description of the various elements and products present in the nature like the edible fruits and vegetables, cereals, meat, different liquids like water, milk and urine with their medicinal properties. It tells us about the innumerable curative properties which mother nature has in store and how to use them for the benefit for our health and live a better life, free from diseases and ailments. The work is in thirteen chapters: (1) *Rasaya*, etc, (2) *Santi*, etc, (3) *camphor*, etc, (4) *gold*, etc, (5) *rats*, etc, (6) *grapes*, etc, (7) *vegetables*, (8) *drinks*, (9) *sweets*, (10) *cereals*, (11) *preparations of food*, (12) *meat*, and (13) *miscellaneous*. The

book was completed on Thursday 8th January 1374. About 500 ingredients of plant, animal and mineral origin which have medicinal properties have been mentioned and briefly described in the book.

Madana (or Madanapala) was the son of Sadhanana, the Raja of Kashi, an offshoot of Taksasiroyalty of the city of Kashi (modern Kashi town of Meerut District). It was the Taksasi family of Sakadwara dynasty that reigned the land of Kannauj from 1000 to 1100. Kannauj is one of the most ancient places of India having rich cultural heritage. The ancient name of this place is Kampakubja. The Chinese pilgrim Fa-hien, visited Kannauj between 399 and 414 A.D. Another famous Chinese pilgrim, Hsuen Tsang visited Kannauj in 643 A.D. An inscription of the Chakrtya dynasty of Lata, dated 1050 A.D. associates the Rashtrakuta dynasty with Kannauj. Rashtrakuta kings were great patrons of science and arts and were themselves learned. The logician Akalanika Bhutta, author of the 'Prajnavartika', the 'Nyayavivichchaya,' the 'Muktavali' and Mahadevacharya, author of the 'Garbhastanamangraha' were patronised by them. The Rashtrakutas at first lived in the north-western part of India and from there they afterwards migrated to the south. They were later period Aryan migrants to Deccan and Maharashtra.

Thus it is evident that Madanapala and his ancestors have always lived in various regions of north and north-western India and were also keen in promoting arts and sciences. At least till 1374, when the writing of Madanapala Nighantu was completed by him (or under his patronage), Madanapala and his ancestors did not even have the remotest connection with the land of Malabar or its people and the indigenous medicinal plants there. Although their gross ecological, climatic and edaphic characteristics differ, being parts of the same tropical land, some common plants naturally occur in the north and north-western India and Malabar.

Susrut Brahmins and Mahanighantu

Charaka, is considered as the author of the book Charaka Samhita, an ancient book on medicine. It is believed to be the oldest of the three ancient treatises of Ayurveda. Some people say that Charaka was not a single individual but several persons of different times who contributed to the knowledge recorded in the book. The name Charaka as well as the title of the book Charaka Samhita reveals its true origin. The word Charaka is a masculine noun of the root word 'char,' which means, "to move" or "to wander". Charaka was a blacksmith and the physician of the emperor Kanishka. Charaka Samhita is studied and considered even today as an authoritative Ayurvedic text on internal medicine in ancient India.

Traditionally orthodox Brahmins consider medical profession as a very lowly and demeaning occupation, as it is forbidden for them by Manu, the law maker of the Hindus. According to the "Laws of Manu", enumerated in Manu Smriti (advised to be followed by Hindus of all stations and castes), the six functions or jobs listed for a Brahmin, in their order of priority are- (1) teaching, (2) studying, (3) sacrificing for himself, (4) sacrificing for others, (5) making gifts, and (6) receiving gifts. Among the above six occupations, the most commendable one is teaching of the Vedas. (Ref: Bahler, 1888, Creapotions, in: Manu Smriti, X: 74-76; 80).

Amongst Hindus, according to the Laws of Manu, practicing the medical profession, or attending to the sick or learning and practicing the art of healing is the job of lowly mixed caste called Ambastha who are primarily Barbers. An Ambastha is an offspring unfavourably begotten by a Brahmin man in association of a lower caste woman (Ref: Mixed Caste, in: Manu Smriti, X: 8). Medical profession, or caring for the sick is not listed among the duties of the higher caste, who are not expected to take up such work that violate their scriptures. (Although Manu Smriti is expected to be followed by all orthodox Hindus without fail, some Sanskrit Brahmins claim that they are at the present day following Parasara Smriti, claiming that in the present times, fourth age of Kali Yuga, it is more relevant).

However, in Buddhism caring for the sick is one of the indispensable duties mentioned for every Buddhist, particularly Buddhist monks. Establishing public hospitals by ancient Buddhist kings is for this purpose, which is considered as a noble act. Buddhist monks living in their monasteries are encouraged to learn about local indigenous medicinal plants so that they can be of help to sick people.

Sanskrit Brahmins claim to be the direct descendants of the migrants from the ancient Sanskrit valley civilization. Till just a few dozen years back (till the last half of 20th century), their social life was inseparable from their own temples. They never bothered to integrate with people outside their community preferring to maintain their traditional way of life, in a cloistered existence. They converse among themselves in their own language only which is not Sanskrit but a simplified different form of it. It is admitted by themselves that "Cochin GSB priests are only competent in performing pujas, none of them are known to be Vedic scholars or Sanskrit scholars. This deprived them of any linguistic skills" (See above).

Till a couple of decades back this was the general condition of the Konkani Brahmins of Cochin, which has not undergone any radical change even today. The Sanskrit Brahmins are very interesting people with many excellent qualities like honesty, integrity and reliability, being God fearing people. But, for want of availability of better men, being Brahmin priests

standing to the religious duties of performing pujas in their temples, the Konkani Brahmins Ranga Bhat, Vinayaka Pandit and Apu Bhat may have doubled-up as healers to the local Konkani, but had predominantly special expertise in (Ayurvedic) medical profession, or any special knowledge about the native medicinal plants of Malabar. Whatever plant ingredients they were using for healing were as ingredients for performing pujas and religious rituals and not on the basis of any separate system of a medical science. (See also the part on witchcraft and miracles of village healers of central Europe mentioned in Chapter 4).

The plants mentioned in the book *Mahonighandu* of Madanapala (written in A.D.1378 in the Kingdom of Kannauj), mentions about 500 ingredients, of which about 400 are plants, while others listed as having therapeutic value are of animal and mineral origin. The plants mentioned by Madanapala are those found growing in various regions of north and north-western India, as he and his ancestors were living there. They did not even have the remotest connection with the land of Malabar or its people or the indigenous medicinal plants of Malabar. Naturally, being of the same tropical region of the world, some common plant species are found in both the regions.

It should also be enlightening that several of the valuable medicinal plants mentioned in *Mahonighandu* can grow only at higher altitudes and would be unable to grow naturally in Malabar. For example, "*Brahma Kamal*" (*Noumea ceylanica*), a rare medicinal plant mentioned in *Mahonighandu*, is a species native to the Himalayas, India, Northern Burma and South-West China. In India it is found only at an altitude of around 4500 m in the Himalayas. Among other valuable medicinal plants mentioned in it, "*Tatamanshi*" (*Hemioclathys grandiflora*) grows at an altitude of about 4500 m, "*Nimbari*" (*Acetyrum onchoceras*) at an altitude of about 4000 m, and "*Demadara*" (*Cedrus deodora*) above an altitude of about 3000-3500 m etc. Among the few non-indigenous drugs mentioned, "*Balakari*" (*Prunus amygdalus*), etc. are mentioned.

Therefore, while *Mahonighandu* mentions a total of about 400 plants only, which are mostly found in regions other than Malabar, any claim that the book *Hortus Malabaricus* which mentions about 341 indigenous medicinal plants growing naturally in Malabar was written based on *Mahonighandu*, is wrong and incredible.

Ranga Bhat, Vinayaka Pandit and Apu Bhat

It is therefore obvious that the knowledge of the three Konkani Brahmins about medicinal plants is mainly restricted to what is mentioned in *Mahonighandu*. When the work on *Hortus Malabaricus* originally began

by Van Rheede the names of the three Brahmins Ranga Shast, Vinayaka Pandit and Apu Shast were recommended to him as helpers in his team, perhaps by the King of Cochín Vira Kerala Varma. In those days of strict caste hierarchy observed by the King, recommending Brahmins alone for this work is understandable. The Konkani Brahmins were also available for constant consultation, as they were living in their Sanskritic locality attached to their temple situated just a stone's throw away from the Dutch Fort at Cochín where Governor Van Rheede was residing. Medical experts among the other local caste Hindus were perhaps unavailable (See: Caste Duties and Occupations in J. Maria Smoot above). The affected population of Konkani, whenever needed, were subjected by their priests to religious rituals or pujas rather than by any qualified medical professional. This was somewhat similar to the condition in medieval Europe where the common people was largely depending upon religious healers and rituals at a time when the science of medicine was not well developed there, and the expensive plant medicine imported through the Arab traders was available only to the few rich living in the port cities.

The Konkani were not familiar with the Malayalam names of the local medicinal plants. They knew only the Sanskrit names as given in *Mohorighanda*, which created the misunderstanding among the Westerners that the language spoken and written in Nagari script by the Konkani was Sanskrit. The separate Konkani names of some of the plants were also known to them, as can be seen written in *Hortus Malabaricus*. The team of St. Mathew, Ranga Shast, Vinayaka Pandit and Apu Shast entrusted the job of collecting the selected medicinal plants from Malabar, by giving their Sanskrit names mentioned in *Mohorighanda*, to the servants (plant collectors). Van Rheede (1682) in his Preface to Volume 3 has recorded: "But still many plants have not been traced, either because he (St. Mathew) had not noted down the names accurately, or the venerable old man had received them from men who did not know the common names of the plants by which the botanists of Malabar designate each plant, or because they were plants from another climate (growing not in Malabar but in native localities of plants mentioned in *Mohorighanda*) (Rheede 1682)".

When Rheede began his work on *Hortus* in 1674-1675, much time had not lapsed after the Konkani came and settled in Cochín. Being Brahmins whose occupation was religious duties in temples, and not supposed to be caring for the sick by their own religious laws (Maria Smoot), their familiarity with the local medicinal plants could, at the most, have only been cursory. Their surnames also confirm that they were temple priests. Shast are priests conducting pujas in smaller temples, while Pandits or Vaidhyas are those exclusive priests performing the rites

for families). Likewise, in the certificates printed in Hortus it is mentioned that the Malabar, Itty Achuden is a well known physician whose father, grand father and great grand father were also physicians or Doctors (Ref. Certificates in Vol. I of Hortus Malabaricus 1678) who were practicing as physicians for centuries. Cheppu or Ezhavas (Itty Achuden was an Ezhava as specifically stated in the certificates) originally were Buddhists for whom caring for the sick is a duty or responsibility as ordained by their religious principles for over two millennia (see Notes above). Claiming that this unique, age-old traditional indigenous medical heritage of Malabar found recorded in Hortus Malabaricus was supplied by the recent migrant Konkani Saraswat Brahmins plants Kanga (that, Vinayaka Parakkal and Apa (that will be, to put it mildly, far from truth. (The above is related not to undervalue the work and the mentions of the three virtuous Brahmins but rather on the contrary).

Independent authors and observers like Cook (2007) have discussed and remarked that: "The factual information contained in Hortus Malabaricus came largely from a variety of indigenous practitioners, especially non-Brahmin healers such as Itty Achuden, the Ezhava. The Hortus Malabaricus provides the best record of the indigenous medical knowledge of the early modern peoples of Malabar, hence its political importance to India as 'prior art' in the current climate of rapacious bio-prospecting". (Harold J. Cook, 2007).

3. Itty Achuden

In the initial period Van Rheede was depending on Fr. Mathew, and the three Konknis, for the information on the therapeutic properties of the medicinal plants being selected. For the drawings of the plants to be included in his book, the actual plant specimens were needed, as Fr. Mathew could supply only crude sketches with writing pens, made from his memory. (See above). The names of the tentatively selected plants were supplied to their servants (plant collectors), by Fr. Mathew and the three Konkis but many of them could not be collected or located as the names supplied to them were not the common names of the plants by which they are known in Malabar, or because they were plants from another climate that grow in localities not of Malabar (Rheede, 1682). As he would not have asked the collectors to bring some plants not found in Malabar or give the names of plants which are unheard of by the people of Malabar, it is evident that at that time Itty Achuden was not involved with the work on Hortus Malabaricus. Itty Achuden would not have wasted his time and energy and that of others, shattering his own reputation by asking the plant collectors to bring from the land of Malabar, specimens of medicinal plants that grow only in the Himalayas.

The Tryst with Itty Achudan

The frustrated Van Rheeke was quick to agree and accept the advice of Paul Herman. "With modest kindness, he (Herman) saw it fit to give us to understand that in this way it would hardly become a book of any importance" (Rheeke, 1882). Therefore Rheeke decided to begin the whole enterprise anew by writing the book by himself. But, he quickly realized that he had no time or expertise to undertake such a technical venture. Rheeke therefore resolved to make the new beginning by recruiting local Malabar experts, who will decide what are the plants to be selected for inclusion in the book, provide their Malayalam names as known to the local people, to make available authoritative information about the diseases to treat, which they are used by the expert physicians of Malabar and the method of preparation and application of the drug by them. Rheeke asked for help from Malabar physicians for this (Heringer, 1988: 145).

His enquiries to locate such an expert Malabar physician to assist him in his proposed work led him to the name of a physician of the name Itty Achudan, who was considered by many as the most famous physician of Malabar in those days. He was popularly known as Collette or Collett Nattayan, the chief of Collett family, a traditional family of ancient physicians of Cardanapalli, a town near Cherthala, about 40 kms away from Cochin. He was the present Head of the family of hereditary physicians who in general had a history of about two thousand years of tradition. Due to the above reasons



Fig. 1. Itty Achudan, Ukkamam drawn by Balan.
(Courtesy: Dr. R. Chandrasekharan)

and the following, it is clear that by the expression "Malabar Physician" inducted above by him was only Ity Achuden alone: (i) Ity Achuden was the only Malabar among the technical experts inducted ahead, (ii) Ity Achuden was the only known professional physician among the new team, (iii) the three Konkani who were essentially temple priests were usually specified by Rheede not by their names but as "Brahmins", (iv) the three Konkani were already associated with the work that was already ongoing on *Hortus Malabaricus*, but discarded in the final second phase.

Rheede had a new strategy for the second beginning of the work on *Hortus Malabaricus*. The Indian scholars should decide which are the plants to be included in the book. They are selected and classified based on the names known to the Malabar physicians. To collect the plants from the Malabar region they (not their servants) are to help. They are to provide the detailed medicinal information on the concerned plants (Heniger, 1986; Fournier, 1967). It should be noted that by the term Indian scholars, Rheede has unambiguously excluded Fr. Mathew. Although he has used the plural word, there is no authentic record to show that any scholars other than Ity Achuden were brought in to participate in the new team. (Ref. comments above, on Konkani Brahmins).

Although Fr. Mathew was excluded from the second plan to write *Hortus Malabaricus*, the three Brahmins Ranga Bhag, Vinayaka Pandit and Appa Bhag were retained in the new team, perhaps because of their alleged expertise in Sanskrit. Their main role was to refer to the Sanskrit book *Maharaghanda* and check whether anything is mentioned in it about the Malabar plants selected by Ity Achuden, and to add if anything more about those plants is known to them. An important point worth noting in this respect is that *Maharaghanda* is a book exclusively on medicine, which mentions about 400 plants that grow in various regions of north and north-western India. On the other hand, *Hortus Malabaricus* gives a detailed illustrated account of 741 plants naturally growing in Malabar alone that are not only of medicinal value but also are of importance for meeting other various essential needs for the people and the country (eg. Ship building). As we know, one of the objectives of Rheede in compiling this 12-Volume treatise was to provide clear scientific proof to his superiors in The Hague that the land of Malabar is self sufficient in all essential raw materials as well as medicines, needed for normal life of the people, without depending on any supply from outside (even under a possible siege). Not only the raw materials but unqualified expertise is also available indigenously.

From all the above facts, it becomes clear that Ity Achuden had a pivotal role to play in providing all the information on the medicinal

plants compiled in *Herbarium Malabaricum*. Nevertheless, there is absolutely no record of him in any historical document available, not only in Kerala but in the whole of India. Even the name of this man is not found to be mentioned anywhere in any context whatsoever. However, in order to recognize the important role of Mr. Schuade, it is not necessary for us to depend on any additional external proof, because it is unequivocally recorded in four out of the six certificates prominently printed in the first pages of the First Volume of *Herbarium Malabaricum* itself. However much one may try to distort or misinterpret the facts, there remain as an undisputable proof on this issue.

For the authenticity of what is recorded in the book, and in order that readers of the book will not mistrust the statements as something provided by someone who knows nothing about medicine or plants, as alleged by the critics of Rhede in VOC including his superior, six Certificates are printed in the first pages of the First Volume of the book. It may be noted that these are printed in First Volume published in 1678 itself, before such allegations against Rhede were ardently raised which were answered by him in 1686 in the Preface of Volume 3. (Hortiger, 1986).

1. Certificate written by Immanuel Camiers, in Malayalam Aryamazha script

The first Certificate is provided by the official interpreter of the Dutch Company in Cochin, Immanuel Camiers, written in his own hand-writing in Malayalam Aryamazha script, is reproduced below:



Fig.4. Certificate by Immanuel Camiers in Malayalam Aryamazha script, [*Herbarium Malabaricum*, Vol. 1, 1678]

I, Immanuel Carneiro, a Roman Christian, native of the City of Cochín, Interpreter under oath, of the Indian Society, as per the order of Governor Henry A. Rivecourt, explained and interpreted from the Malabar language into Portuguese, the names, medical powers and properties of trees, plants, herbs and creepers growing in these Malabar territories (as heard from Colatida, a Malayalee-gentile Doctor and of Chego (Bhagwat) race, a native of Canapuran, and written in the book of Malabar Doctors; in testimony of which I wrote these and signed. Given in the City of Cochín, the 19th day of April of the year 1673.

MANUEL CARNEIRO,

Translated from Portuguese language by me

CHRISTIAN HERMAN de GONAP

Civil Secretary of the City of Cochín.

2. Certificate given by Immanuel Carneiro in Latin

The second certificate is written by Immanuel Carneiro in Portuguese language and translated by Christian Herman de Gonap, Civil Secretary of the City of Cochín, from Portuguese into Latin.

An English translation (Manila, 2003) of it is given below.

This is

I, Immanuel Carneiro, a Roman Christian, native of the City of Cochín, Interpreter under oath, of the Indian Society, as per the order of Governor Henry A. Rivecourt, explained and interpreted from the Malabar language into Portuguese, the names, medicinal powers and properties of trees, plants, herbs and creepers growing in these Malabar territories (as heard from Colatida, a Malayalee-gentile Doctor and of Chego race, a native of Canapuran, and written in the book of Malabar Doctors; in testimony of which I wrote these and signed. Given in the City of Cochín, the 19th day of April of the year 1673.

Manuel Carneiro

Translated from Portuguese by me

Christian Herman de Gonap

Civil Secretary of the City of Cochín.

4. Another Certificate by Itty Achuden in Latin

A fourth-certificate by Itty Achuden, which is translated from Malayalam into Portuguese and then from Portuguese into Latin, is also provided.

An English translation (Manilal, 2003) of it is given below:

This is

I, Itty Achuden, a Malabari Doctor, Chigo by race, gentle and native of Canaparam or the place called Coddle Canapalli, Inhabitant of the house called Colabolla, who was born of great-grand parents, grant parents and parents who were physicians or Doctors, testify that I came to the City of Cochim as per the order of Governor Henry A. Rheade, and through Manuel Carneiro, Interpreter of the Noble Indian Society, told and dictated names, medicinal powers and properties of plants, trees, herbs and creepers, written and explained in our book and which (plants and their medicinal powers) I had observed by long experience; that this explanation and dictation went on without any doubt, nor would any of the Malabari doctors doubt about the veracity of the things I said, I made those which I wrote by my own hand and signed.

Given in the City of Cochim, 28 April, 1675.

ITTY ACHUDEN,

Malabar Doctor.

Translated from the Malayalam language into Portuguese by

MANUEL CARNEIRO,

And from Portuguese into Latin by me

CHRISTIAN HERMAN de BOMER,

Civil Secretary of the City of Cochim

5. Certificate by three Konkani written in Konkani language, in Nagari script

The fifth certificate is written in Konkani language and signed jointly by Ranga Bhat, Vinayaka Pandit and Apa Bhat together. It is not mentioned in whose hand-writing the certificate is written but it is believed that it was Vinayaka Pandit who wrote that.

An English translation of the above Konkani certificate is given below:

This is

Savarni Sree Salivahana Saka 1597, Ashvina Samvatsara Chaitra Bahula 10th. We, Ranga Bhat, Vinayaka Pandit and Apa Bhat, three

ॐ नमो भगवते वासुदेवाय ॥ १ ॥
 ॐ नमो भगवते वासुदेवाय ॥ २ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ३ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ४ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ५ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ६ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ७ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ८ ॥
 ॐ नमो भगवते वासुदेवाय ॥ ९ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १० ॥
 ॐ नमो भगवते वासुदेवाय ॥ ११ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १२ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १३ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १४ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १५ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १६ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १७ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १८ ॥
 ॐ नमो भगवते वासुदेवाय ॥ १९ ॥
 ॐ नमो भगवते वासुदेवाय ॥ २० ॥

ॐ नमो भगवते वासुदेवाय ॥

ॐ नमो भगवते वासुदेवाय ॥

ॐ नमो भगवते वासुदेवाय ॥

Fig.6. Joint Certificate in Konkani language, given by Ranga Bhat, Vinayaka Pandit and Apu Bhat.

physicians residing in City of Cochín upon direction from Julian Van
 Rheede, Commodore of Cochín, had seen knowledgeable persons,
 giving payment, to respective places of Malabar and brought the
 respective medicinal creepers, plants, roots, fruits and seeds that grow
 there in respective seasons to draw their sketches. Afterwards we have
 given the names and various qualities of those medicines as described
 in our medical dictionaries and as found in our long experience, and we
 have personally supervised the compilation of this manuscript for the
 past two years morning and evening. Not to regard this as labor we have
 put our signatures. To authenticate this as truth we have given this in
 writing in Nagari Script.

Ranga Bhat Sd/. Vinayaka Pandit Sd/. Apu Bhat Sd/.

6. Certificate by the three Konkani given in Latin

The sixth certificate is also a joint certificate signed together by
 Ranga Bhat, Vinayaka Pandit and Apu Bhat. This is the Latin version of a
 certificate translated from "the language of the Brahmins" [i.e. Konkani]
 by Vinayaka Pandit into Portuguese language, and from Portuguese into
 Latin by Christian Herman de Deenop.

The English version of that Latin certificate, reproduced here is rendered by Manilal (2004).

This is

From the year 1555, called (named) by Brahmins Pottanapottu (which with us is the year 1624) to the 20th of April of the year called Requetan (which with us is 1675), We, Ranga Bhatt, Vinayaka Pandit and Appa Bhatt, all three from the race and religion of Brahmins and ancient Gymnosophists, in the District of Cochin, as per the order of Noble Lord Henry, a Pihanda, Governor of Malabar Territories and of the City of Cochin, through our servants who have knowledge of trees, plants, herbs and crops, caused them to be brought from the Malabar region with their flowers, fruits and seeds to the above mentioned city, in order that they may be drawn and described, the names of which are written in our book called **MAHARATHOLATTU** in which are contained their medicinal powers and properties and besides these, we have added to these whatever we have learned and observed by long experience and with great labour and sweat and in order that this book may be completed, we have worked for the space of almost two years, so that each and anybody who will read or see it, could have faith in it which he himself would have given, if he himself would have experienced the things written in it; we swear therefore according to our custom that all that is contained in it is true, in testimony of which we made these papers (documents) which we have signed with our own hands, Given in the city of Cochin, on the 20th day of April, 1675.

Ranga Bhatt, Vinayaka Pandit, Appa Bhatt.

Translated from the language of the Brahmins by the undersigned

VINAYAKA PANDIT

into the Portuguese language,

and from Portuguese into Latin by me

CHRISTIAN HERRMAN de SCHIEP

Civil Secretary of the City of Cochin.

The above Certificate/Statements by those who were associated with the book are given prominently in the first pages of the First Volume of Hortus Malabaricus, to confirm the authenticity of the book and to show the original source of information given in it. The certificates are evidence that Wende can not be accused of plagiarism or stealing from other peoples' works. He wanted to let the readers of the book know the truth regarding the matter.

It is note-worthy in this connection, that the first two certificates about the contributions by Ity Achuden, are signed and given by Manuel Carneiro, Official Interpreter of the Dutch Company in Cochin,

and Christian Herman de Sonog, Civil Secretary of Cochín VOC. This amounts to official attestation of the contribution of Iry Achuden. No such attestation of the contributions of Konkani is found to be included. Apart from the certificates/statements given by the Konkani themselves, their names are not mentioned by the Company Officials anywhere.

Of the above six certificates, the last two are given jointly by the Konkani Brahmins- Ranga Shet, Vinayaka Pandit and Apa Shet. They were involved in the work of *Hortus Malabaricus*, along with Fr. Mathew during the (about) two years, 1674-1675. However, it has been shown that all the work turned out during that period were discarded as useless and unacceptable. The work produced by the Konkani during that period also naturally would have been included in that bundle. It has been clarified earlier that *Mohonighandu*, the book being consulted as an authoritative Lexicon by the Konkani, was written by others on the plants growing in other regions and climates. Therefore, as mentioned earlier, it can be found that *Mohonighandu* could have been only of limited use in the writing of *Hortus Malabaricus*.

In their certificates it is mentioned that the Konkani asked their servants to collect from Malabar, the selected plants mentioned in *Mohonighandu*, by telling the names of plants to them. The main occupation of these Konkani who had migrated to Cochín only comparatively recently, was conducting proper worship in their temples. It is but natural, therefore, that their knowledge on the indigenous medicinal plants of Malabar was limited. Hence it can be found that the main work assigned to them was to see whether anything is mentioned about the selected plants in the Sanskrit book *Mohonighandu* and supplement what is known about them, if any. That their job was only as supplementary or supporting staff is also clear from the following three reasons. (1) Whatever were produced by them earlier, when they were working with Fr. Mathew were later and finally rejected. (2) Their reference dictionary *Mohonighandu* is a book containing information about only 400 plants, that occur in a totally different geographic and climatic region. Apart from some common species, the important indigenous plants found growing naturally in Malabar are not mentioned in it. (3) In his certificates, Inmanuel Carrero, the **official interpreter under oath** of the Dutch Company has nowhere mentioned the names of the Konkani Brahmins. On the other hand, it is clearly certified in the two statements signed by Carrero that all the information obtained by him was provided by Iry Achuden (Colakida). In the first certificate given by him he has written under oath that 'the names, medical powers and properties of trees, plants, herbs and creepers growing in these Malabar territories (as) heard from Colakida, a Molapole gentle Doctor and of

Chega (Echeval race, a native of Caragunam, and written in the book of Malabari Doctors'. In the second certificate also he has certified that what is recorded by him are: the names, medicinal powers and properties of trees, plants, herbs and creepers growing in these Malabari territories (as) heard from Colabida, a Molakke-gentile Doctor and of Chega race, a native of Caragunam, and written in the book of Malabari Doctors'. Noteworthy and significant points from the above are that (1) the plants in question are those actually growing in Malabari territories, and not in any other far-away regions, (2) the information is provided by Ity Achuden, Colabida who is a professional Doctor, (3) the information is recorded in the book of Malabari Doctors and not taken from any exotic books.

In the third and fourth certificates printed in *Hortus Malabaricus*, provided by Ity Achuden, the present head of the Colabida family who is entitled to use the designation (Title) Collet Nidyan (Collet Physician), Ity Achuden has certified in the third certificate that "The methods of treatment with them (drug plants) etc. for various diseases, each separately and as described in (our) ancestral books and as known from practical experience. All these are told and explained thus, to the Interpreter of the Company, Immanuel Carneiro", and in the fourth certificate that "through Manuel Carneiro, interpreter of the Noble Indian Society, told and dictated names, medicinal powers and properties of plants, trees, herbs and creepers, written and explained in our book and which (plants and their medicinal powers) I had observed by long experience; that this explanation and dictation went on without any doubt, nor would any of the Malabari doctors doubt about the veracity of the things I said".

From all the above it is unmistakably clear that it was the Malabari Doctor Ity Achuden who provided the information about the indigenous medicinal plants of Malabar and their therapeutic powers as utilized by the people of Malabar during the past very many centuries, that are described in *Hortus Malabaricus*. This medicinal tradition of Malabar is a remnant of the 2500 year-old Buddhist tradition of its early ancient people of the land, which is curiously found to be maintained even today.

As is understandable, many persons must have been involved during various steps and stages of the compilation and production of such a comprehensive, mammoth and expensive work. However, the book was conceived, planned and put together facing immense difficulties at every stage, by the Dutch Governor of Cochin, Hendrik Adriaan Van Rheede, even when he had to face dismissal from his coveted and powerful post by his superiors in the VOC who could not fully understand or appreciate

the value of the academic pursuit of a soldier-administrator, Van Rheeke undeniably continued with his project on *Hortus Malabaricus* till its completion. The original manuscripts/books from which the hereditary medical knowledge of the ancient people of Malabar were taken, and the knowledge of the native scholars who preserved this traditional knowledge for over two millennia are now irrecoverably lost. *Hortus Malabaricus* is the only authentic record of this. Therefore, had Van Rheeke not taken up and published this work in print, an important chapter in the cultural history and heritage of Malabar, nay this part of India, would have vanished for ever.

No wonder, long after the great empires and granite fortresses built by the mighty sword of many a successive European power have crumbled to dust and shrunk into a few desiccated lines in some moldy books of history, this empire of knowledge, built by the simple quill of a far-sighted Dutch soldier and celebrated as *Hortus Malabaricus*, remains unconspicuously respected and glittering even today.

Hortus Malabaricus and the Socio-Cultural Heritage of India

This is a collection of essays about the indigenous medical knowledge of the early modern people of Malabar and their ancient culture, mainly based on *Hortus Malabaricus*, the first ever printed book on the flora of India written in Latin and published during 1613 - 1625 from Amsterdam. It was compiled and edited by the then Dutch-Governor of Cochin, Commendator Hendrik Adriaen van Rheenke based on the information supplied by 15 native physicians (Lakshy Achudan/Kadige) on the important indigenous medicinal plants, the diseases to treat while they are used and the methods of preparation of the drugs and treatments.

The original book in 12 volumes was the result of the first ever collaboration venture between European scholars in the Netherlands and Indian experts of Malabar in Asia, the like of which did not happen again for centuries. This book is now used to protect the indigenous medical knowledge on about 700 Indian medicinal plants from patenting by others, protecting their traditional knowledge as the intellectual property of the local people themselves. Apart from its traditional medical, botanical and academic importance, the book may be found to be intimately associated with the socio-cultural legacy and the ancient heritage of India, as revealed in the maps/pennanulations.



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Indian Association for Angiosperm Taxonomy (IAAT)